Measurable functions

REAL VARIABLES I

Fall 2026

In this section we investigate when a function is compatible with an σ algebra structure defined on its domain and codomain.

For a function $f\colon X \to Y$ and $B \subseteq Y$ define

$$f^{-1}(B) \coloneqq \{x \in X : f(x) \in B\}.$$

If X and Y carry σ-algebras, measurability requires every measurable set in the target space to pull back to a measurable set in the domain.

After studying these notes, you will be able to do the following.

- Verify measurability by testing inverse images of a generating family.
- Construct pullback, product, and trace σ-algebras.
- Apply level-set criteria to real- and extended-real-valued functions.
- Prove that measurable functions remain measurable under pasting, algebraic operations, lattice operations, and pointwise limits when the operations are defined.

6.1 Measurable maps

Definition 6.1 (Measurable map). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A function $f\colon X \to Y$ is $(\mathcal{A}, \mathcal{B})$-*measurable* if

$$f^{-1}(B) \in \mathcal{A} \qquad \text{for every } B \in \mathcal{B}.$$

When the σ-algebras are fixed, we say that f is *measurable*.

The definition is expressed through inverse images because inverse images preserve the set operations used in a σ-algebra. For every family $(B_i)_{i\in I}$ of subsets of Y,

$$f^{-1}(Y) = X, \qquad f^{-1}(Y \setminus B) = X \setminus f^{-1}(B),$$
$$f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i), \qquad f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i).$$

These identities hold for arbitrary unions and intersections, not only for countable ones.

Note that, in general for two sets $A_1, A_2 \in \mathcal{A}$ one merely has $f(A_1 \cap A_2) \subset f(A_1) \cap f(A_2)$, where $f(A) \coloneqq \{f(x) \in Y : x \in A\}$. Other desireable identities also fail.

Proposition 6.2 (Composition of measurable maps). Let $(X, \mathcal{A}) \xrightarrow{f} (Y, \mathcal{B}) \xrightarrow{g} (Z, \mathcal{C})$ be measurable maps. Then $g \circ f\colon (X, \mathcal{A}) \to (Z, \mathcal{C})$ is measurable.

Proof of Proposition 6.2. The inverse image of a set under a composition is an iterated inverse image. For $C \in \mathcal{C}$, measurability of g gives $g^{-1}(C) \in \mathcal{B}$. Measurability of f then gives

$$(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C)) \in \mathcal{A}.$$

□

The inverse images of all measurable target sets themselves form a σ-algebra.

Definition 6.3 (Pullback σ-algebra). Let $f\colon X \to Y$ be a function and let $\mathcal{B}$ be a σ-algebra on Y. The *pullback σ-algebra* is

$$f^{-1}(\mathcal{B}) \coloneqq \{f^{-1}(B) : B \in \mathcal{B}\}.$$

Proposition 6.4 (Minimality of the pullback). The collection $f^{-1}(\mathcal{B})$ is a σ-algebra on X. It is the smallest σ-algebra on X for which $f\colon X \to (Y, \mathcal{B})$ is measurable.

Proof of Proposition 6.4. The inverse-image identities verify the σ-algebra axioms, and the definition of measurability then gives minimality. The inverse-image identities give

$$X = f^{-1}(Y) \in f^{-1}(\mathcal{B}),$$

and show that $f^{-1}(\mathcal{B})$ is closed under complements and countable unions. Thus it is a σ-algebra.
If $\mathcal{A}$ is any σ-algebra on X for which f is measurable, then $f^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}$. Hence $f^{-1}(\mathcal{B}) \subseteq \mathcal{A}$, which proves the minimality assertion. □

6.1.1 Checking a generating family

To check that a mapping is measurable, it is not necessary to check Definition 6.1 over every member of the target σ-algebra. It is enough to check measurability of the pullbacks for a family of sets that generates the target σ-algebra.

Proposition 6.5 (Generator criterion for measurability). Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces, and suppose that $\mathcal{G} \subseteq 2^Y$ satisfies $\mathcal{B} = \sigma(\mathcal{G})$. A function $f\colon X \to Y$ is measurable if and only if

$$f^{-1}(G) \in \mathcal{A} \qquad \text{for every } G \in \mathcal{G}.$$

Proof of Proposition 6.5. The reverse implication collects all target sets with measurable inverse image into a σ-algebra. The forward implication follows from $\mathcal{G} \subseteq \mathcal{B}$. For the reverse implication, define

$$\mathcal{H} \coloneqq \{E \subseteq Y\colon f^{-1}(E) \in \mathcal{A}\}.$$

The inverse-image identities show that $\mathcal{H}$ is a σ-algebra on Y. The hypothesis gives $\mathcal{G} \subseteq \mathcal{H}$, and therefore

$$\mathcal{B} = \sigma(\mathcal{G}) \subseteq \mathcal{H}.$$

Thus $f^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{B}$. □

Example 6.6 (Constant functions). Every constant function $f\colon (X, \mathcal{A}) \to (Y, \mathcal{B})$ is measurable. Indeed,

if $f(x) = y_0$ for every $x \in X$, then

$$f^{-1}(B) = \begin{cases} X, & y_0 \in B, \\ \emptyset, & y_0 \notin B. \end{cases}$$

Both possible inverse images belong to $\mathcal{A}$.

Example 6.7 (Indicator functions). For $E \subseteq X$, define the indicator function

$$\mathbb{1}_E(x) \coloneqq \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

Give $\{0,1\}$ its full σ-algebra. Then $\mathbb{1}_E \colon (X, \mathcal{A}) \to \{0,1\}$ is measurable if and only if $E \in \mathcal{A}$. The nontrivial implication follows from

$$E = \mathbb{1}_E^{-1}(\{1\}).$$

For the reverse implication, every inverse image under $\mathbb{1}_E$ is one of $\emptyset$, E, $X \setminus E$, or X. The same equivalence holds when $\mathbb{1}_E$ is viewed as an $\mathbb{R}$-valued function: the inverse image of a Borel set depends only on its intersection with $\{0,1\}$.

6.1.2 Product measurable spaces

Definition 6.8 (Product σ-algebra). For measurable spaces $(Y, \mathcal{B})$ and $(Z, \mathcal{C})$, their product σ-algebra is

$$\mathcal{B} \otimes \mathcal{C} \coloneqq \sigma(\{B \times C \colon B \in \mathcal{B},\ C \in \mathcal{C}\}).$$

Example 6.9 (Coordinate projections). The coordinate projections

$$\pi_Y \colon (Y \times Z, \mathcal{B} \otimes \mathcal{C}) \to (Y, \mathcal{B}), \qquad \pi_Z \colon (Y \times Z, \mathcal{B} \otimes \mathcal{C}) \to (Z, \mathcal{C})$$

are measurable. For $B \in \mathcal{B}$ and $C \in \mathcal{C}$,

$$\pi_Y^{-1}(B) = B \times Z, \qquad \pi_Z^{-1}(C) = Y \times C,$$

and both sets belong to $\mathcal{B} \otimes \mathcal{C}$.

Proposition 6.10 (Measurability of a paired map). Let $f \colon (X, \mathcal{A}) \to (Y, \mathcal{B})$ and $g \colon (X, \mathcal{A}) \to (Z, \mathcal{C})$, and define

$$(f, g)(x) \coloneqq (f(x), g(x)).$$

Then $(f, g) \colon (X, \mathcal{A}) \to (Y \times Z, \mathcal{B} \otimes \mathcal{C})$ is measurable if and only if f and g are measurable.

Proof of Proposition 6.10. The two directions use the rectangle generators and the coordinate projections, respectively.

Step 1: test the paired map on measurable rectangles.
Suppose first that f and g are measurable. For every measurable rectangle $B \times C$,

$$(f, g)^{-1}(B \times C) = f^{-1}(B) \cap g^{-1}(C) \in \mathcal{A}.$$

The measurable rectangles generate $\mathcal{B} \otimes \mathcal{C}$, so Proposition 6.5 applies.

Step 2: recover the coordinate maps by projection.
Conversely, if (f, g) is measurable, then

$$f = \pi_Y \circ (f, g), \qquad g = \pi_Z \circ (f, g).$$

The coordinate projections are measurable by Example 6.9; hence Proposition 6.2 gives the conclusion. □

For Euclidean spaces, $\mathcal{B}(\mathbb{R}^m) \otimes \mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R}^{m+n})$. One inclusion follows because rectangles with open factors are Borel. For the other inclusion, open rectangles with rational endpoints form a countable basis for $\mathbb{R}^{m+n}$, so every open set is a countable union of such rectangles. Consequently, a map $F\colon X \to \mathbb{R}^d$ is measurable if and only if each of its coordinate functions is measurable.

6.1.3 Borel and Lebesgue measurability

For a topological space Y, write $\mathcal{B}(Y)$ for the σ-algebra generated by the open subsets of Y.

Theorem 6.11 (Continuous maps are Borel measurable). Let X and Y be topological spaces. Every continuous map

$$f\colon (X, \mathcal{B}(X)) \to (Y, \mathcal{B}(Y))$$

is measurable.

Proof of Theorem 6.11. The generator criterion reduces the claim to inverse images of open sets. The open subsets of Y generate $\mathcal{B}(Y)$. If $V \subseteq Y$ is open, continuity gives that $f^{-1}(V)$ is open in X, and hence $f^{-1}(V) \in \mathcal{B}(X)$. Apply Proposition 6.5. □

Definition 6.12 (Borel and Lebesgue measurable functions). A function $f\colon \mathbb{R}^d \to \mathbb{R}$ is

1. *Borel measurable* if it is $(\mathcal{B}(\mathbb{R}^d), \mathcal{B}(\mathbb{R}))$-measurable;
2. *Lebesgue measurable* if it is $(\mathcal{L}(\mathbb{R}^d), \mathcal{B}(\mathbb{R}))$-measurable, where $\mathcal{L}(\mathbb{R}^d)$ is the Lebesgue σ-algebra.

Since $\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d)$, every Borel measurable real-valued function on $\mathbb{R}^d$ is Lebesgue measurable. Careful, the converse does not hold (since the inclusion $\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{L}(\mathbb{R}^d)$ is actually strict)!

6.1.4 Level-set criteria

For a real-valued function, measurability can be checked using only super or sub-level sets.

Proposition 6.13 (Level-set criterion for real-valued functions). Let $(X, \mathcal{A})$ be a measurable space and let $f\colon X \to \mathbb{R}$. The following conditions are equivalent:

1. f is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$-measurable;
2. $\{x \in X : f(x) > a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
3. $\{x \in X : f(x) \geq a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
4. $\{x \in X : f(x) < a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$;
5. $\{x \in X : f(x) \leq a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.

In condition 2, it is enough to use $a \in \mathbb{Q}$. The same rational restriction can be made in conditions 3–5.

Proof of Proposition 6.13. The proof first identifies one countable generating family of rays and then passes between the four inequality conventions.

Step 1: generate the Borel sets from rational upper rays.
The rays (q, ∞), $q \in \mathbb{Q}$, generate $\mathcal{B}(\mathbb{R})$. To verify this assertion, first observe that

$$(a, \infty) = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} (q, \infty).$$

Also,

$$[b, \infty) = \bigcap_{\substack{q \in \mathbb{Q} \\ q < b}} (q, \infty), \qquad (-\infty, b) = \mathbb{R} \setminus [b, \infty).$$

Thus every open interval belongs to the σ-algebra generated by these rational rays. Every open subset of $\mathbb{R}$ is a countable union of open intervals with rational endpoints, so the rational rays generate all of $\mathcal{B}(\mathbb{R})$. The equivalence of conditions 1 and 2 now follows from Proposition 6.5.

Step 2: pass between strict and non-strict level sets.
The remaining choices of inequality are equivalent by complements and countable approximations. For example,

$$\begin{aligned}
\{f \geq a\} &= \bigcap_{n=1}^{\infty} \{f > a - 2^{-n}\}, \\
\{f < a\} &= X \setminus \{f \geq a\}, \\
\{f \leq a\} &= X \setminus \{f > a\}, \\
\{f > a\} &= \bigcup_{n=1}^{\infty} \{f \geq a + 2^{-n}\}.
\end{aligned}$$

The rational versions follow from the same generator argument. □

The distinction between strict and non-strict inequalities matters when a set identity is formed. For example, if $h(x) = \sup_{n \in \mathbb{N}} f_n(x)$, then

$$\{h \leq a\} = \bigcap_{n \in \mathbb{N}} \{f_n \leq a\}, \qquad \{h > a\} = \bigcup_{n \in \mathbb{N}} \{f_n > a\}.$$

In contrast, the condition $f_n(x) < a$ for every n does not force $\sup_n f_n(x) < a$.

6.2 Operations on real-valued measurable functions

Theorem 6.14 (Algebra of real-valued measurable functions). Let $f, g \colon (X, \mathcal{A}) \to \mathbb{R}$ be measurable and let $\alpha \in \mathbb{R}$. Then the following functions are measurable:

1. $f + g$, $f - g$, fg, and αf;
2. $\max\{f, g\}$, $\min\{f, g\}$, and $|f|$;
3. the positive and negative parts

$$f^+ \coloneqq \max\{f, 0\}, \qquad f^- \coloneqq \max\{-f, 0\};$$

4. f/g on the measurable subspace $\{x \in X : g(x) \neq 0\}$.

If f/g is assigned any fixed real value on $\{g = 0\}$, the resulting function on X is also measurable.

Proof of Theorem 6.14. The proof establishes addition by a countable level-set identity, derives the algebra and lattice operations, and treats division on its measurable domain.

Step 1: prove measurability of addition from rational level sets.
We first prove measurability of $f+g$ directly from level sets. For $a \in \mathbb{R}$, density of $\mathbb{Q}$ gives the identity

$$\{f+g<a\} = \bigcup_{\substack{(q,r)\in\mathbb{Q}^2 \\ q+r<a}} (\{f<q\} \cap \{g<r\}).$$

If $f(x)<q$, $g(x)<r$, and $q+r<a$, then $f(x)+g(x)<a$. Conversely, if $f(x)+g(x)<a$, choose positive $\varepsilon < a-f(x)-g(x)$. Density of $\mathbb{Q}$ supplies $q, r \in \mathbb{Q}$ such that

$$f(x)<q<f(x)+\varepsilon/2, \qquad g(x)<r<g(x)+\varepsilon/2.$$

Then $q+r<a$. The displayed union is countable, and each of its terms is in $\mathcal{A}$, so $f+g$ is measurable by Proposition 6.13.

Step 2: derive the remaining algebra and lattice operations.
The function $t \mapsto -t$ is continuous, hence Borel measurable by Theorem 6.11. Composition shows that $-g$ is measurable, and therefore $f-g=f+(-g)$ is measurable. The same argument with $t \mapsto \alpha t$, $t \mapsto t^2$, and $t \mapsto |t|$ proves measurability of αf, f^2, and $|f|$. The identity

$$fg = \frac{(f+g)^2 - f^2 - g^2}{2}$$

then proves measurability of the product.
The formulas

$$\max\{f,g\} = \frac{f+g+|f-g|}{2}, \qquad \min\{f,g\} = \frac{f+g-|f-g|}{2}$$

give the maximum and minimum. Taking $g=0$ and using measurability of $-f$ gives f^+ and f^-. In particular,

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Step 3: handle division on its natural domain.
The set $D := \{g \neq 0\}$ is measurable by Proposition 6.13. The restrictions of f and g to D, where D has the trace σ-algebra, are measurable, and $g(D) \subseteq \mathbb{R} \setminus \{0\}$. Since $t \mapsto 1/t$ is continuous on $\mathbb{R} \setminus \{0\}$, the function $1/g$ is measurable on D, and so is $f/g = f(1/g)$. Finally, $X \setminus D = \{g=0\}$ is measurable. Pasting f/g on D with a constant function on $X \setminus D$ proves the last assertion. □

The proof of addition uses rational parameters because a σ-algebra is closed under countable unions, not arbitrary unions. The pairing result gives another proof: $(f,g)\colon X \to \mathbb{R}^2$ is measurable and addition, multiplication, maximum, and minimum are continuous maps from $\mathbb{R}^2$ to $\mathbb{R}$.

Analogous statements hold for extended-real-valued functions.

6.2.1 Pointwise limits

Theorem 6.15 (Countable lattice operations and pointwise limits). Let $(f_n\colon X \to \bar{\mathbb{R}})_{n\in\mathbb{N}}$ be a sequence of measurable functions. Then the following extended-real-valued functions are measurable:

1. $\sup_{n\in\mathbb{N}} f_n$ and $\inf_{n\in\mathbb{N}} f_n$;
2. $\limsup_{n\to\infty} f_n$ and $\liminf_{n\to\infty} f_n$.

Consequently, if $f_n(x) \to f(x)$ in $\bar{\mathbb{R}}$ for every $x \in X$, then f is measurable.

Proof of Theorem 6.15. The proof handles suprema and infima by level sets, builds limsup and liminf from tail operations, and then identifies an existing pointwise limit.

Step 1: prove measurability of countable suprema and infima.
Left as an exercise.

Step 2: form limsup and liminf from tail operations.
For every $N \in \mathbb{N}$, the tail supremum and tail infimum

$$S_N \coloneqq \sup_{n \geq N} f_n, \qquad I_N \coloneqq \inf_{n \geq N} f_n$$

are measurable by the first part. The identities

$$\limsup_{n \to \infty} f_n = \inf_{N \in \mathbb{N}} S_N, \qquad \liminf_{n \to \infty} f_n = \sup_{N \in \mathbb{N}} I_N$$

then prove the second part.

Step 3: identify a pointwise limit.
If $f_n(x) \to f(x)$, then at every $x \in X$,

$$f(x) = \liminf_{n \to \infty} f_n(x) = \limsup_{n \to \infty} f_n(x).$$

Hence f is measurable. □

Even if every f_n is real-valued, a countable supremum, infimum, limsup, or liminf can take an infinite value. The codomain $\bar{\mathbb{R}}$ makes each of these operations defined without an additional boundedness hypothesis.

Example 6.16 (A discontinuous pointwise limit). On $[0, 1]$, let $f_n(x) = x^n$. Every f_n is continuous and hence Borel measurable. Pointwise,

$$f_n(x) \longrightarrow \mathbb{1}_{\{1\}}(x).$$

The limit is discontinuous at 1, but it is measurable by Theorem 6.15. Thus pointwise limits preserve measurability even though they need not preserve continuity.

Warning (Uncountable suprema). Countability in Theorem 6.15 cannot be removed. Left as an exercise.

Problem. Let $(f_n : X \to \mathbb{R})_{n \in \mathbb{N}}$ be a sequence of measurable functions. Show that the set of $x \in X$ such that $\lim_n f_n(x)$ exists is measurable.